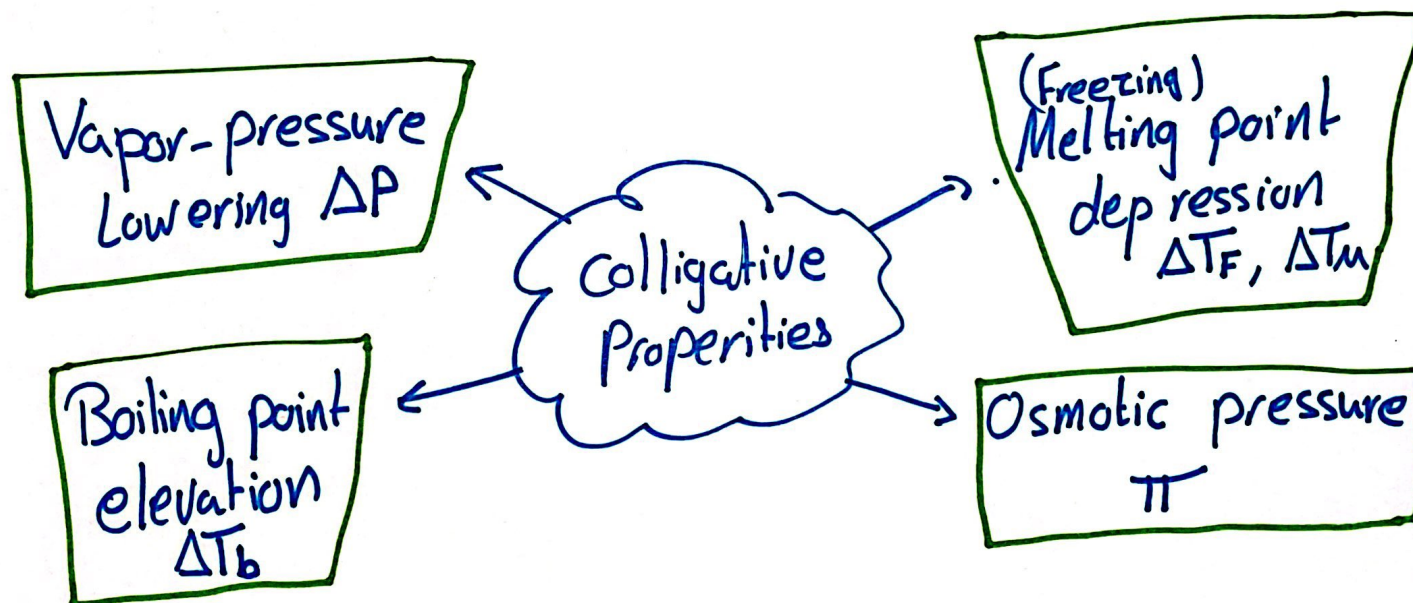


Colligative Properties

- properties that depend on the number (concentration) of solute particles presents in solution, not_{on} the identity or nature of the solute particles.

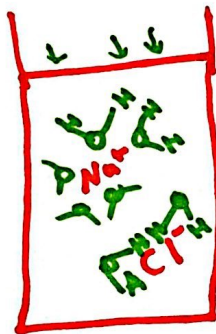
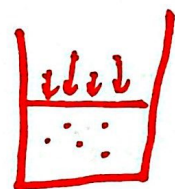


Vapor Pressure Lowering

Raoult's Law

$$P_{\text{solution}} = \chi_{\text{solvent}} P_{\text{solvent}}^{\circ}$$

$$\chi_{\text{solvent}} = \frac{n_{\text{solvent}}}{n_{\text{solvent}} + n_{\text{solute}}}$$



NaCl

$P_{\text{solvent}}^{\circ}$
Vapor pressure
of pure solvent

P_{solution}
Vapor pressure
of solvent
over solution.

$$X_{\text{solvent}} + X_{\text{solution}} = 1 \Rightarrow X_{\text{solution}} = 1 - X_{\text{solvent}}$$

$$X_{\text{solvent}} = 1 - X_{\text{solution}}$$

$$\Delta P = P_{\text{solvent}}^{\circ} - P_{\text{solution}}$$

$$= P_{\text{solvent}}^{\circ} - X_{\text{solvent}} P_{\text{solvent}}^{\circ}$$

$$= P_{\text{solvent}}^{\circ} (1 - X_{\text{solvent}})$$

$$= P_{\text{solvent}}^{\circ} \cdot X_{\text{solution}}$$

amount of solute $\uparrow$ ✓
mole fraction solvent $\downarrow$

vapor pressure of solvent over solution $\downarrow$

✓ ΔP
vapor pressure lowering $\uparrow$

What is the vapor pressure of water at 35°C over a solution made by mixing of:

[A] 30 g Urea with 180 g Water at 35°C

$$- n_{\text{urea}} = \frac{30}{60} = 0.5 \text{ mol}$$

$$- n_{\text{H}_2\text{O}} = \frac{180}{18} = 10 \text{ mol}$$

$$- X_{\text{soln}} = \frac{10}{10+0.5} = 0.952$$

$$\begin{aligned} - P_{\text{H}_2\text{O}}^{\text{soln}} &= X_{\text{soln}} P_{\text{H}_2\text{O}}^{\circ} \\ &= 0.952 \times 42.18 \\ &= \boxed{40.17 \text{ torr}} \end{aligned}$$

At $35^{\circ}\text{C} \Rightarrow$

Vapor pressure of pure water is 42.18 torr.

Mm urea = 60 g/mol

Mm H_2O = 18 g/mol

$$\begin{aligned} \Delta P &= P_{\text{soln}}^{\circ} - P_{\text{soln}} \\ &= 42.18 - 40.17 \\ &= \boxed{2.01 \text{ torr}} \end{aligned}$$

[B] 60 g urea with 180 g H_2O at 35°C ?

$$- n_{\text{urea}} = \frac{60}{60} = 1$$

$$- n_{\text{H}_2\text{O}} = 10 \text{ mol}$$

$$- X_{\text{solvent}} = \frac{10}{11} = 0.909$$

$$\begin{aligned} - P_{\text{soln}} &= X_{\text{solv}} P_{\text{solv}}^{\circ} \\ &= 0.909 \times 42.18 \\ &= \boxed{38.34} \end{aligned}$$

$$\begin{aligned} \Delta P &= P_{\text{solv}}^{\circ} - P_{\text{solution}} \\ &= 42.18 - 38.34 \\ &= \boxed{3.84} \end{aligned}$$

□ 90 g urea with 180 g water at 35°C?

$$n_{\text{urea}} = \frac{90}{60} = 1.5$$

$$X_{\text{solv}} = \frac{10}{10 + 1.5} = 0.87$$

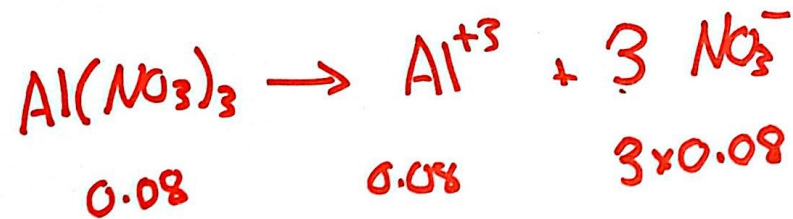
$$\begin{aligned} P_{\text{solv}} &= X_{\text{solv}} P_{\text{solv}}^{\circ} \\ &= 0.87 \times 42.18 \\ &= \boxed{36.68 \text{ torr}} \end{aligned}$$

$$\begin{aligned} \Delta P &= P_{\text{solv}}^{\circ} - P_{\text{solv}} \\ &= 42.18 - 36.68 \\ &= \boxed{5.5 \text{ torr}} \end{aligned}$$

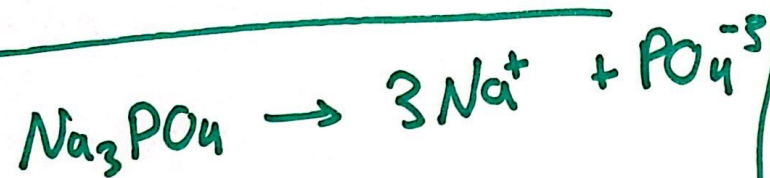
Which of the following aqueous solution have the highest vapor Pressure lowering?

	n	m	$n \times m$
[A] 0.08 m $\text{Al}(\text{NO}_3)_3$	4	0.08	0.32
[B] 0.1 m NaCl	2	0.1	0.2
[C] 0.04 m Na_3PO_4	4	0.04	0.16
[D] 0.08 m Na_2SO_4	3	0.08	0.24
[E] 0.1 m sucrose.	1	0.1	0.1





$$\begin{array}{l} 4 \times 0.08 \\ = \underline{0.32} \end{array}$$



Boiling point Elevation

$$T_{\text{soln}} > T_{\text{solu}}$$

$$\Delta T_b = K_b \cdot m$$

$$\text{molality} = \frac{n_{\text{solute}}}{\text{kg solven}}$$

$T_{\text{soln}} - T_{\text{solu}}$

molal B.P
elevation
constant.

$$K/m \rightarrow K \cdot \text{kg/mol}$$

$$C/m \rightarrow K \cdot \text{kg/mol}$$



- Solvent dependent

amount of solute ↑

n_{solute} ↑

molality ↑

boiling point elevation ↑

B.P soln ↑

Freezing Point Depression

$$\Delta T_F =$$

$$T_{\text{solu}} - T_{\text{soln}}$$

$$K_F \cdot m \rightarrow \text{molality}$$

molality
freezing depression
constant K_F $\rightarrow$ kg/mol
 C $\rightarrow$ kg/mol

$$K_F \cdot m \rightarrow C$$

Schroeder dependent.

amount of solute $\uparrow$
molality $\uparrow$

$$\Delta T_F \uparrow$$

$$T_F \text{ Solu} \downarrow$$

What is the Freezing point of a solution containing 478g of ethylene glycol ($C_2H_6O_2$, 62g/mol) in 3202g of Water?

$$K_F = 1.86 \text{ } ^\circ\text{C}/m.$$

$$\begin{aligned}\Delta T_F &= K_F \cdot m \\ &= 1.86 \times 2.41 \\ &= 4.48 \text{ } ^\circ\text{C}\end{aligned}$$

$$\Delta T_F = T_{\text{solv}} - T_{\text{solut}}$$

$$4.48 = 0 - T_{\text{solut}}$$

$$T_{\text{solut}} = -4.48 \text{ } ^\circ\text{C}$$

$$C + 273.15 = K$$

$$\begin{aligned}n_{\text{solute}} &= \frac{478}{62} \\ &= 7.71 \text{ mol}\end{aligned}$$

$$\begin{aligned}\text{molality} &= \frac{7.71}{3.202} \\ &= \underline{\underline{2.41 \text{ m}}}\end{aligned}$$

What is the Molecular weight of adrenaline if 0.64 g of adrenaline dissolved in 36 g of CCl_4 raises the boiling point by 0.49°C ? $K_b[\text{CCl}_4] = 5.02^\circ\text{C/m}$

$$\Delta T = K_b \times m$$

$$0.49 = 5.02 \cdot m$$

$$m = 0.0976 \text{ m}$$

$$m = \frac{n_{\text{solute}}}{\text{kg solvent}}$$

$$0.976 = \frac{n}{36 \times 10^{-3}}$$

$$n = 3.5 \text{ mole} \times 10^{-3}$$

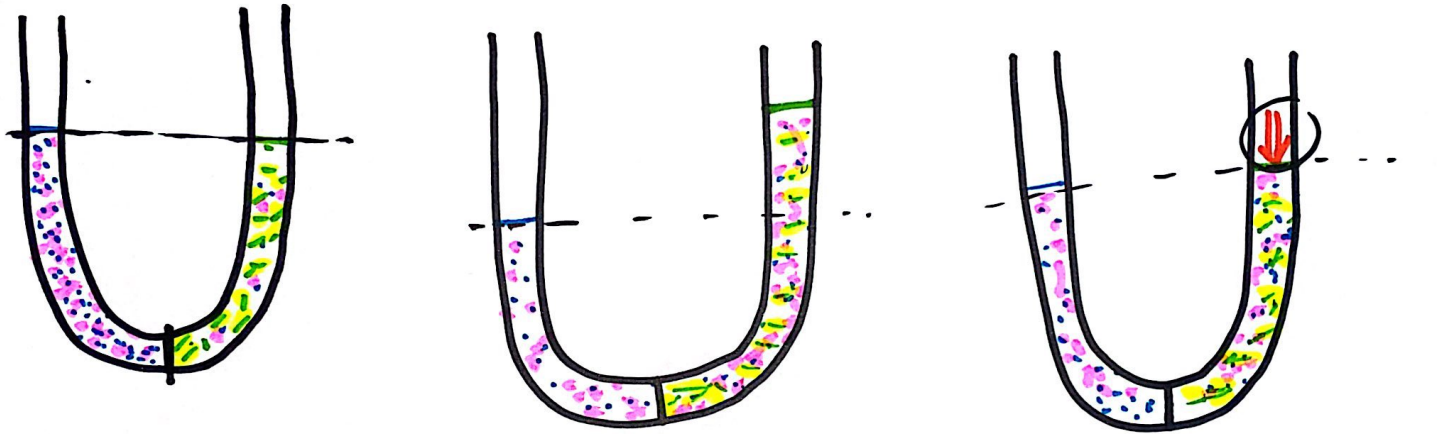
$$n = \frac{\text{mass}}{\text{M.M}}$$

$$3.5 \times 10^{-3} = \frac{0.64}{\text{MM}}$$

$$\text{MM} = \frac{0.64}{3.5 \times 10^{-3}}$$

$$= 182.86 \text{ g/mol}$$

Osmosis



~~movement of~~
movement of solvent from the area of higher solvent
concentration to area with lower solvent concentration

$$PV = nRT$$

Osmotic Pressure $[\pi]$



the pressure required to stop osmosis.

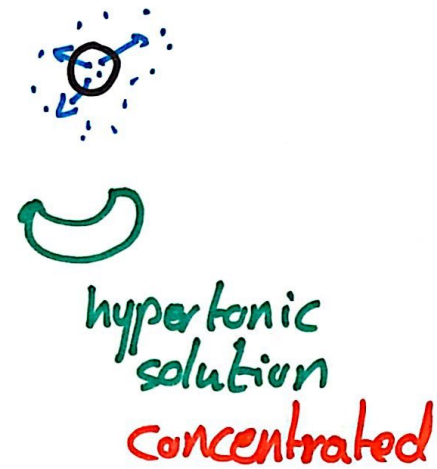
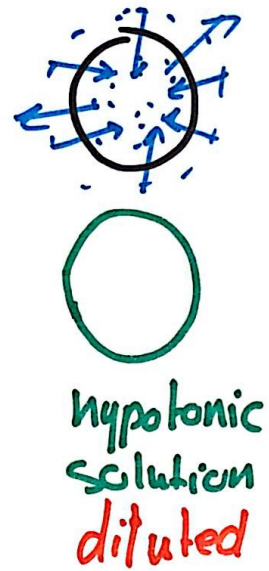
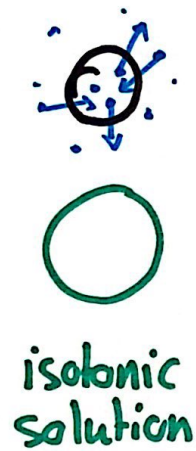
$$\pi V = nRT$$

$$\pi V = \frac{\text{mass}}{MM} RT \Rightarrow \pi = \frac{(\text{mass})}{MM V} RT$$

$$MM = \frac{mRT}{\pi V}$$

$$\pi = \frac{mRT}{MM V}$$

In a cell :



At 298 K, 25 ml of solution containing 27.55 mg of protein has an osmotic pressure of 3.22 torr. The Molecular weight of protein is

$$\pi V = nRT$$

$$4.24 \times 10^{-3} \times 25 \times 10^{-3} = n \times 0.0821 \times 298$$
$$n = 4.33 \times 10^{-6}$$

$$n = \frac{m}{MM}$$

$$MM = \frac{m}{n} = \frac{27.55 \times 10^{-3}}{4.33 \times 10^{-6}}$$

$$= \boxed{6363.6 \text{ g/mol}}$$

$$1 \text{ atm} = 760 \text{ torr}$$

$$X = 3.22$$

$$X = 4.24 \times 10^{-3} \text{ atm}$$

Which of the following aqueous solution will have the highest osmotic pressure?

[A] 1m $\text{Ca}(\text{NO}_3)_2$	3	3
[B] 3m KCl	2	6
[C] 2.5m MgS	2	5
[D] 4m $\text{C}_6\text{H}_{12}\text{O}_6$	1	4
[E] H_2O	0	0

Chapter 13 ✓

